Unit: 6 Unit 8: Quadratics **Name:** ____________
Topic: Graphing Quadratics and Properties of Parabolas **Date:** ________

Challenge (Habanero)

Q1. A chef is designing a parabolic roof for a kitchen. The equation of the parabola is $y = x^2 + 4x + 4$. What is the vertex of the parabola?
(A) $(0, 0)$
(B) $(-2, 0)$
(C) $(-2, -4)$
(D) $(-2, 4)$
(E) $(0, 4)$

Q2. A construction worker is building a bridge with a parabolic shape. The equation of the parabola is $y = -x^2 + 6x - 8$. What is the axis of symmetry of the parabola?
(A) $x = -1$
(B) $x = 1$
(C) $x = 2$
(D) $x = 3$
(E) $x = 4$

Q3. A recipe for a cake calls for x cups of flour, where x satisfies the equation $y = x^2 - 4x + 4$. What is the minimum value of y?
(A) 0
(B) 1
(C) 2
(D) 4
(E) 8

Q4. An architect is designing a parabolic dome for a building. The equation of the parabola is $y = 2x^2 - 4x + 1$. What is the y-intercept of the parabola?
(A) $(0, 0)$
(B) $(0, 1)$
(C) $(0, 2)$
(D) $(0, 3)$
(E) $(0, 4)$

Q5. A construction company is building a parabolic tunnel. The equation of the parabola is $y = -2x^2 + 8x - 7$. What is the maximum value of y?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q6. A chef is making a parabolic sauce dispenser. The equation of the parabola is $y = x^2 + 2x - 6$. What is the x-intercept of the parabola?
(A) $(-3, 0)$
(B) $(-2, 0)$
(C) $(-1, 0)$
(D) $(1, 0)$
(E) $(2, 0)$

Q7. An engineer is designing a parabolic antenna. The equation of the parabola is $y = -x^2 + 2x + 5$. What is the direction of the parabola?
(A) Up
(B) Down
(C) Left
(D) Right
(E) Horizontal

Q8. A contractor is building a parabolic staircase. The equation of the parabola is $y = 2x^2 - 8x + 9$. What is the vertex form of the parabola?
(A) $y = 2(x - 2)^2 + 1$
(B) $y = 2(x + 2)^2 - 1$
(C) $y = 2(x - 1)^2 + 2$
(D) $y = 2(x + 1)^2 - 2$
(E) $y = 2(x - 3)^2 + 3$

Open Ended Questions (FRQ)

Q9. A chef is designing a parabolic kitchen counter. The equation of the parabola is $y = x^2 - 6x + 9$. Find the vertex, axis of symmetry, and y-intercept of the parabola. Show all work and explain your answers.

Q10. An architect is designing a parabolic roof for a building. The equation of the parabola is $y = -x^2 + 4x - 3$. Find the maximum value of y and the x-intercepts of the parabola. Show all work and explain your answers.

Chef's Tips

- Tip 1: Make sure to check the units of the answers.
- Tip 2: Use the vertex form of a parabola to find the vertex and axis of symmetry.
- Tip 3: Use the quadratic formula to find the x-intercepts of a parabola.